PRACTICE WORKSHEET – COMPUTER SCIENCE (CS)

Class – XII

Topic 1: Python Fundamentals & Data Types

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1. State whether the following statement is **True** or **False**:

The output of the following code will be 6.0

```
import math
print(math.sqrt(36))
```

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2. Which of the following is a **mutable** data type in Python?

a) String

b) Tuple

c) Dictionary

d) int

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3. Which of the following is **not** a valid Python data type?

a) complex

b) long

c) float

d) bool

Topic 2: Operators & Expressions

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4. What will be the output of the following code?

```
print(25 > 15 and 10 < 5 or not 3 > 1)
```

a) True

b) False

c) Error

d) No output

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5. What will be the output of the following Python statement?

```
print(18 + 2 ** 3 - 24 // 4)
```

---

6. Which of the following operators is used for **membership** testing in Python?

a) is

b) ==

c) in

d) not

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7. Write the output of the following code:

```
x = 8
y = 12
print(x < y and y > 10 or x == 8)
```

Topic 3: Strings

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8. What will be the output of the following Python code?

```
text = "HelloWorld"
print(text[-5:-2])
```

a) Wor

b) rld

c) oWo

d) llo

---

9. What is the output of the given Python code?

```
message = 'Red-Green-Blue'
print(message.split("-"))
```

a) ['Red', 'Green', 'Blue']

b) ['Red-Green-Blue']

c) ['Red', 'Green-Blue']

d) Error

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10. Predict the output of the following code:

```
sentence = "Learning Python is enjoyable"
print(sentence.partition("is"))
print(sentence.count("n"))
```

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11. Write a statement to convert the string "welcome to python" to **title case**.

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12. Write a statement to find the index of the first occurrence of the substring "code" in a string named message.

Topic 4: Lists & Tuples

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13. What will be the output of the following code?

```
colors = ["red", "blue", "green", "yellow", "purple"]
print(colors[2][0] + colors[-2][-1])
```

a) gw

b) gy

c) yl

d) gp

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14. Predict the output of the following code:

```
data = (100, 200, 300, 400, 500)
print(data[1:4])
```

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15. Write a statement to sort the elements of list items = [50, 20, 60, 10, 30] in **descending order**.

Topic 5: Dictionaries

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16. What will be the output of the following Python code?

```
student = {"name": "Amit", "age": 18, "grade": "A"}
print(student.get("subject", "Not Available"))
```

a) None

b) Not Available

c) Error

d) Amit

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17. What will be the output of the following code?

```
record = {'a': 5, 'b': 10, 'c': 15}
record['d'] = 20
print(record.keys())
```

Topic 6: Loops (for & while)

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18. Write the output of the following Python code:

```
for num in range(3, 18, 4):
  print(num, end=" ")
```

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19. Predict the output of the following code:

```
i = 5
while i < 20:
  print(i, end="-")
  i = i + 5
```

---

20. What will be the output of the following code?

```
for m in range(1, 9, 2):
  if m == 5:
    continue
  print(m, end=" ")
```

Topic 7: Functions & Scope

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21. What will be the output of the following Python code?

```
value = 7
print(value, end="..")
def update():
   global value
   value = value + 4
   print(value, end="//")
update()
print(value)
```

a) 7..11//11

b) 7..7//11

c) 7..11//7

d) 11..11//11

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22. Differentiate between:

- I. **Local variable** and **Global variable**
- II. list() and tuple() functions

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23. Write a Python function calculate_product() that accepts a list of numbers as an argument and returns the **product** of all numbers in the list.

Example: If the list is [2, 3, 4], the function should return 24.

Topic 8: Conditional Statements

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24. What will be the output of the following Python code?

```
a = 12
b = 8
c = 15
print(a > b and c > a or not b > c)
```

a) True

b) False

c) Error

d) No output

Topic 9: Importing Modules

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25. Which of the following will correctly **import** only the floor and sqrt functions from the math module?

a) import math

b) from math import *

c) from math import floor, sqrt

d) import floor, sqrt from math

Topic 10: Random Module

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26. What possible output is expected from the following code?

```
import random
numbers = [8, 16, 24, 32, 40]
start = random.randint(1, 2)
end = random.randint(3, 4)
for pos in range(start, end + 1):
  print(numbers[pos], end="%")
```

a) 16%24%

b) 16%24%32%

c) 24%32%

d) 8%16%24%

Topic 11: Error Correction

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27. The code below is intended to **extract the last three characters** of a string and return them. Rewrite it after correcting all errors. Underline the corrections made.

```
define get_last_three(str)
  if len(str) < 3:
    return str
  result = str[-3:-0]
  return result
output = get_last_three("Computer")
Print("Last three chars:" output)
```

Topic 12: Functions with Lists & Dictionaries

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28. A. Write a function delete_value() in Python that accepts a list items and a number target. If the target exists in the list, remove its **last occurrence**. If it does not exist, print "Value not present".

B. Write a Python function add_product() that accepts a dictionary inventory, a product_id, and a stock. The function should add the product id and stock to the dictionary. If the product id already exists, print "Product already in inventory" instead of updating it.

Topic 13: String & List Traversal

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29. A. Predict the output of the following Python code:

```
text = "DATA-456"
output = ""
index = 0
```

```
while index < len(text):
    if text[index] >= '0' and text[index] <= '9':
        digit = int(text[index])
        digit = digit * 3
        output = output + str(digit)
    elif text[index] >= 'A' and text[index] <= 'Z':
        output = output + text[index].lower()
    else:
        output = output + '@'
    index = index + 1
print(output)
```

B. Predict the output of the following Python code:

```
countries = ["India", "France", "Germany", "Brazil", "Canada"]
selected = []
for country in countries:
    if country[0] in 'CDEFG':
        selected.append(country.upper())
print(selected)
```

Topic 14: List Operations (Max, Min, Sum, Frequency)

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30. Write a Python program to perform the following operations on a list of integers:

- **I.** Create a list of 8 integers (you may hardcode or take input).
- **II.** Find and print the **maximum**, **minimum**, and **sum** of the numbers.
- **III.** Count and print the **frequency** of a specific number (ask the user which number to search for) using linear search.

Example: If the list is [3, 7, 3, 9, 12, 3, 15, 18] and the user searches for 3, the output should be Frequency of 3 is 3.

Topic 15: Dictionary Operations

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31. A. Write a Python program to:

- Create a dictionary with names of 5 **students** as keys and their **marks** as values.
- Display all students whose marks are greater than **80**.
- Find the **total marks** of all students.

B. Write a Python program to:

- Create a list of 8 numbers entered by the user.
- Find and display the **maximum**, **minimum**, and **sum** of all numbers.
- Also display the numbers in **ascending order**.

Topic 16: Output Prediction (Mixed)

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32. Predict the output of the following Python code. Write the output for each code block:

```
# Code Block 1
list_a = [10, 20, 30, 40]
list_a.insert(2, 25)
print(len(list_a))

# Code Block 2
tuple_b = (2, 4, 6, 8, 10)
print(tuple_b[-3:-1])
```

```
# Code Block 3
dict_c = {'p': 100, 'q': 200, 'r': 300}
dict_c['s'] = 400
print(dict_c.items())

# Code Block 4
for val in range(10, 50, 10):
  if val == 30:
    break
  print(val, end="-")
```

Topic 17: User-Defined Function with List

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33. A. Write a Python program using a **user-defined function** that accepts a list of numbers as an argument and returns the **product of all positive numbers** in the list. Also write the main program that:

1. Takes 6 numbers from the user and stores them in a list.
2. Calls the function with this list.
3. Prints the product of positive numbers.

   Example: If the list is [2, -3, 4, -1, 5, 6], the output should be Product of positive numbers: 240 (because 2 × 4 × 5 × 6 = 240).

B. Write a Python program that:

1. Takes a sentence (string) input from the user.
2. Counts and displays the number of **uppercase letters** in the sentence.
3. Also displays the number of **lowercase letters** in the sentence.

   Example: Input: "Hello World 123" Output: Uppercase: 2, Lowercase: 8